

Session 23: Research & Build

This is the session where your paper actually gets **BUILT**. You'll read your sources deeply, run your hands-on AI exploration, and draft 4 of the 8 paper sections.



Recap - Where We Left Off

In Session 22 you:

- Picked your domain & sharpened your research question
- Understood the 8-section paper structure
- Found 5–8 verified sources (Perplexity → Gemini → NotebookLM)
- Committed to a hands-on exploration plan

In homework you:

- Filtered + verified your sources
- Wrote one-sentence summaries of each source
- Sketched your Abstract placeholder
- Drafted Introduction prose (100–200 words)

Objectives

By the end of this Session, you'll have accomplished all six of these milestones. Mentor co-builds every section.

01

Deep Source Reading

Read 2–3 sources deeply in class - extract specific findings

03

Methodology Sketch

Sketch your Methodology section (~80–120 words)

05

Source Grouping

Group your sources for Related Work refinement

02

Hands-On Exploration

Complete your hands-on exploration - and document it

04

Results Started

Start your Results section - headline + 2–3 sentences

06

User Testing Plan

Commit to a user testing plan - 3–5 testers identified by name



TWO Threads

Your investigation runs on two parallel tracks. **Both matter equally.** Neither alone is enough. Both together = real research.



LITERATURE THREAD

- Read your sources **DEEPLY**
- Pull out specific findings
- Build the foundation for your Results section



HANDS-ON THREAD

- Use a tool to experience how AI works in your domain
- Watch what it does, where it struggles, what surprises you
- Become the **EVIDENCE** for your investigation

The Deep Literature Read

In S22, you verified sources existed. Today, you READ them properly.

Two-Part Plan

In Class Today

You and your mentor will read **2–3 sources together** to build the muscle

In Homework

You'll read the **remaining 2–5 sources** on your own using the same workflow

For EACH source, extract these 4 things:

Main Finding

What's the source's big takeaway?

Specific Facts

Statistics, examples, cases you can cite

Direct Quotations

Short (under 15 words) only when wording matters

Source Relationships

Does this agree, disagree, or extend others?

❗ Use NotebookLM - it's grounded in source material and far less likely to make stuff up. Still, always check its answers against the actual source.

Reading ONE Source - Step by Step

For each source, follow this exact workflow. With your mentor in class - work through 2–3 sources together. Your mentor will coach you on what "main finding" actually means and how to spot specifics.



Upload Source to NotebookLM

Add your source document directly into NotebookLM to begin the analysis process.



Find the Main Finding

Ask: *"What is this source's main finding about AI in [your domain]?"*



Gather Evidence

Ask: *"What specific examples, statistics, or cases does it provide?"*



Identify Limitations

Ask: *"What limitations or concerns does it raise?"*



Save to Your Notes Doc

Record all answers - you'll cite directly from these notes later in your work.

In homework - apply the same 5 steps to the remaining sources, solo. **Don't skim. Don't trust the summary. Get into the actual content.**

Synthesizing Across Sources

After reading each source individually, look at them **TOGETHER**. This is what SYNTHESIS means. Your Related Work isn't a list - it's a woven pattern.



Where do they **AGREE**?

What do your sources collectively confirm?



Where do they **DISAGREE**?

What tensions or contradictions exist?



Unique Contributions

What does Source A say that Source B doesn't?



Repeated Examples?

Same finding cited in multiple places?

A Pattern to Watch For

Most sources fall into rough groupings. Organize Related Work around these - not source-by-source. Organizing around groupings = your Related Work feels like **REAL research**.



What AI DOES

Sources describing AI applications in this domain



How Well It Works

Sources reporting effectiveness and performance



Concerns

Sources raising bias, safety, and ethics issues



Policy & Regulation

Sources discussing rules and oversight

Mini Activity 1 - Group YOUR Sources!

Together with your mentor, group your 5–8 sources into 2–4 categories. Most sources will fit into 2–3 groups - that's normal. This grouping = the **BACKBONE** of your Related Work section.

Source	What Group?
Source 1: _____	(e.g., "What AI Does")
Source 2: _____	(e.g., "Concerns")
Source 3: _____	(e.g., "How Well It Works")
Source 4: _____	(e.g., "Policy")
Source 5: _____	(e.g., "Concerns")

☑ This grouping = the **BACKBONE** of your Related Work section.

Time for the Hands-On Exploration!

Now the **FUN** part. You actually use AI in your domain.

What this IS:

- A learning activity
- It gives you firsthand experience with how AI works in your area
- Your observations become part of your paper

What this ISN'T:

- Not the MAIN contribution of your paper
- Not "I built a new AI"

 Think of it as: **gathering evidence for your investigation.**

Set Up the Exploration FIRST

Before you touch your tool, **WRITE DOWN** your setup notes. Don't skip this. Writing down your setup = the difference between "research exploration" and "playing around."



Which tool?

Name the specific tool you'll use (Teachable Machine, ChatGPT, ElevenLabs, etc.)



What will I try?

Describe the specific test or task you'll run



What do I PREDICT will happen?

Write your hypothesis before you start



What am I watching for?

Define what success, failure, and surprise look like

While You Explore - Pay Attention

As you do the exploration, **ACTIVELY observe** and take notes. Take notes **AS YOU GO**. Don't try to remember after.

✅ What does the tool do well?

Note specific successes with concrete examples

❌ Where does it struggle or fail?

Document failures precisely - what broke and when

😮 Does anything surprise you?

Unexpected results are often the most valuable findings

🤝 Does it match your sources?

Compare live observations to what you read - or contradictions

Examples of Hands On Explorations

See what real explorations look like. Your exploration goes in this same shape.

Domain	What Student Does	What to Watch For
Food allergies	Train Teachable Machine on peanut/no-peanut food photos	Where it works, where it fails, confidence vs. accuracy
Music education	Use ElevenLabs to generate same sentence in 5 emotions	Are emotions distinct? Can listeners ID them?
Wildlife ID	Try iNaturalist on 15 local plants/animals	Which species confident? Which wrong? How handles unclear photos?
Math practice	Use ChatGPT to generate problems at different grade levels	Quality, errors, difficulty matching grade

Hands-On - RUN IT NOW!

Time to actually do your exploration.

01

Open your tool

Teachable Machine / ChatGPT / ElevenLabs / etc.

02

Run your test as planned

Follow the setup you wrote down before starting

03

Save screenshots as you go

Name them **Figure 1**, **Figure 2**, etc. - you'll include the most important ones in your final paper

04

Write observations in real time

Don't wait - capture what you see as it happens

👉 This is the heart of today. By the end: you have evidence + screenshots + observations.

Document Your Observations

Right after the exploration, write these down - be **SPECIFIC**. These observations = your Results section + your Bias/Safety analysis.



What worked well

1–2 concrete examples of success



Where it struggled

1–2 concrete failures with specifics



What surprised you

And exactly why it was surprising



What you learned

About AI in your specific domain

- ⊗ Vague = useless. Replace *"it was kinda bad"* with *"it misidentified the peanut sauce 3 out of 5 times even at high confidence."*

Draft Methodology Now

Methodology = how **YOU** investigated. Today, just SKETCH it. Use this 4-part structure.

Brief Restatement

What you were investigating - state your core research question or focus in 1–2 sentences.

Literature Workflow

What sources you used, how you found them, and how you verified their credibility.

Exploration Setup

What tool you used, what you tried, and what you observed or watched for.

Synthesis Note

How you combined the two threads - literature and exploration - into a unified approach.



Target today: ~80–120 words sketch



Homework: expand to 150–250 words

Methodology Example

Here's what a finished Methodology section looks like. Yours should look like this **in YOUR voice**.

"This investigation drew on two threads. For the literature thread, I used Perplexity to find 12 sources about AI in food allergy management, then narrowed to 6 using a credibility checklist. I verified each in NotebookLM. For the hands-on thread, I trained a Teachable Machine classifier to tell peanut foods from non-peanut foods using 15 photos per class. I tested it on 10 new photos. I observed where it worked, where it failed, and whether it matched what my sources described. I synthesized both threads by comparing the classifier's failure patterns to the bias patterns the literature flagged."

- ☑ That's a real Methodology section. Notice: specific numbers, specific tools, specific observations - all in a student's natural voice.

Drafting Results - Start Today

Results = where literature meets your exploration. **SYNTHESIS.**



Key Literature Findings

Key things your literature showed about AI in your domain



Exploration Reveals

What your hands-on exploration revealed firsthand



Where They Confirmed

Where literature and exploration confirmed each other



NEW Discoveries

Where your exploration found something the literature missed



Homework: expand after user testing. Target: 150–250 words

Honesty Reminder

A quick rule about Results: **Include what DIDN'T work.**

If the tool worked imperfectly → say so
Partial success is still honest, valuable data

If something was WORSE than expected → say that
Negative findings often = the MOST interesting discoveries

Bias and safety problems live in the failure cases
The best researchers report what failed - that's where the real insight is

⊗ Don't shape your Results to look impressive. Shape them to be **TRUE**.

Refine Your Related Work

You'll refine Related Work fully in homework. But while we're in class, take a 2-minute pass on the groupings from the earlier activity.

Are you happy with the groups?

Review the categories you assigned your sources to earlier in class

Did today's reading shift anything?

Did anything you read deeply today change how you'd categorize a source?

Move one source if needed

Move one source into a different group if today's reading changed your mind

☑ Small adjustments here = big improvements when you refine Related Work at home.

User Testing - Plan for Homework

For homework: **3–5 people** try YOUR exploration and tell you what they noticed.

Who?

Family, friends, teacher, even a kind neighbor. Don't stress about "diversity" - just get honest reactions from a few different people. If you can include one person who's different from you in some way (age, background, comfort with tech), their reaction is often the most interesting finding.

What you ask:

- "What did the tool seem to do well?"
- "Where did it confuse you?"
- "Did anything surprise you?"
- "Would you trust this for [your domain's purpose]?"
- "What would make you trust it more?"

 For each tester: write down who they are, what they tried, their reactions, patterns across testers.

Outputs Checklist - Did You Hit Everything?

By end of today, check that you have all of these. If anything is missing - flag it now. Don't carry gaps into homework.

✅ In-Class Outputs

- ✅ ~~2–3 sources read deeply in class with mentor scaffolding~~
- ✅ ~~Hands-on exploration completed (with named screenshots: Figure 1, Figure 2...)~~
- ✅ ~~Observations documented (specific, not vague)~~
- ✅ ~~Methodology section sketched (~80–120 words)~~
- ✅ ~~Results section started (headline + 2–3 sentences)~~
- ✅ ~~Sources grouped for Related Work (with any updates from today's reading)~~
- ✅ ~~User testing plan committed – 3–5 testers identified by NAME~~

🏠 Coming in Homework

- ☐ Deep-read the remaining 2–5 sources solo
- ☐ Refine Related Work with all sources read
- ☐ Expand Methodology + Results
- ☐ Run user testing

Your Homework

Four tasks, in this order. Every word in your own voice - mentor will spot AI-generated phrasing.

Deep-read remaining 2–5 sources

Same 5-step NotebookLM workflow from class: Upload → Main Finding → Specific Examples → Limitations → Save notes. Extract specific findings from each.

User Testing

Have 3–5 testers try your exploration. For each: write down who they are, what they tried, their reactions. Look for patterns across testers - these are your strongest findings.

Refine + Expand Three Sections

Related Work : 150-200 words using ALL sources now deeply read.

Methodology : expand sketch to 150-250 words.

Results : expand with user testing + literature synthesis (150–250 words).

Note 3 SURPRISES

What surprised you most about your exploration and user testing? These feed Session 24's Bias & Safety analysis.

 **Voice rule reminder:** Every word in your own voice. Mentor will spot AI-generated phrasing.

Next Time - Session 24:

Analyze & Synthesize

Bring everything to S24: refined Related Work, Methodology, expanded Results, user testing notes, 3 surprises. **By end of S24: paper is essentially complete and presentation is started.** 🎓



Apply the 5 Lenses

Bias, Safety, Ethics, Human-in-the-Loop,
Voice Integrity



Draft Bias & Fairness

Full section written and refined



Draft Safety & Governance

Full section written and refined



Draft Conclusion & Future Work

Wrap up your investigation with forward-
looking insights



Polish All 6 Sections

Final Abstract + AI Attribution Statement



Start Gamma Presentation

First 3 slides of your final presentation